BEYOND THE 60/40 PORTFOLIO

Regime-Sensitive Multi-Asset Allocation for High Net Worth Investors

Evidence Framework for Inflation, Interest-Rate, and Economic-Stress Regimes

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Research objective

To evaluate whether transparent, long-only multi-asset and regime-sensitive portfolios improved downside protection and risk-adjusted performance relative to a traditional 60/40 benchmark in a rolling, out-of-sample test from January 2013 through December 2025.

Code and research materials:
https://github.com/ishaanparm/beyond-60-40-portfolio

WORKING PAPER STATUS
This manuscript reports a completed out of sample analysis with public code, methodology, and archived result outputs.
It has not been peer reviewed and should be cited as an independent working paper. The results are educational and are not investment advice.

Keywords: asset allocation; high-net-worth investing; 60/40 portfolio; risk parity; conditional value-at-risk; inflation regimes; monetary policy; wealth preservation

JEL classification: G11, G12, E31, E43

Abstract

This paper evaluates whether diversified, optimized, and regime-sensitive multi-asset portfolios improved wealth preservation relative to a traditional 60/40 stock-Treasury benchmark for a hypothetical high-net-worth investor. Seven liquid exchange-traded fund proxies were studied using monthly data from January 2008 through December 2025. The main out-of-sample test covers January 2013 through December 2025 and applies quarterly rebalancing, 10 basis points of one-way turnover costs, long-only constraints, and lagged macroeconomic signals. The portfolios include a strategic diversified allocation, constrained maximum-Sharpe mean-variance optimization, equal-risk contribution, minimum 95 percent conditional value-at-risk, and a precommitted inflation-and-policy regime strategy. The 60/40 benchmark delivered the highest nominal CAGR (9.56 percent) and real CAGR (6.70 percent). Mean-variance achieved the highest Sharpe ratio (1.036) and Sortino ratio (1.734), while reducing maximum drawdown to 12.16 percent from 20.56 percent. Minimum CVaR produced the lowest volatility and tail loss but only a 3.99 percent CAGR. The regime-sensitive strategy reduced drawdown and CVaR, yet its 6.80 percent CAGR, weaker Sharpe ratio, and 43.02 percent annual turnover caused it to fail the precommitted success standard. The evidence favors simple strategic exposure for growth and constrained optimization when downside control is the dominant objective.

Executive Summary

The completed analysis produces a clear trade-off rather than a single universally superior portfolio. The 60/40 benchmark was the strongest compounding portfolio, while the constrained mean-variance and minimum-CVaR portfolios were substantially more defensive. The regime-sensitive overlay reduced losses but surrendered too much growth and did not improve risk-adjusted performance.

- Growth result: the traditional 60/40 portfolio earned a 9.56 percent nominal CAGR and a 6.70 percent real CAGR, the highest among the six portfolios.
- Downside result: mean-variance reduced maximum drawdown to 12.16 percent and achieved the strongest Sharpe and Sortino ratios; minimum CVaR delivered the lowest 95 percent CVaR at 2.27 percent.
- Regime result: the regime-sensitive allocation reduced maximum drawdown to 17.44 percent from 20.56 percent, but its real CAGR lagged the benchmark by 2.69 percentage points and its annual turnover reached 43.02 percent.
- Robustness result: transaction costs of 0 to 25 basis points did not change the main ranking, and mean-variance downside protection remained broadly stable across lookback, rebalancing, and weight-cap variants.

Empirical conclusion
From 2013 through 2025, the traditional 60/40 portfolio produced the strongest nominal and real growth. Mean-variance and minimum-CVaR portfolios materially reduced volatility, drawdowns, and tail losses, but at the cost of lower long-run returns. The precommitted regime-sensitive strategy improved downside protection but did not justify its complexity under the paper's success standard.

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1. Introduction

Asset allocation is the central portfolio decision in private wealth management because it determines how a client is exposed to economic growth, inflation, interest rates, credit conditions, liquidity risk, and market drawdowns. Security selection can matter, but a portfolio constructed without an explicit allocation framework can produce unintended concentrations even when it contains many individual holdings. The practical challenge is therefore not to identify a single “best” asset. It is to combine assets whose return drivers differ enough to support the client’s long-term objective under several economic environments.

The 60/40 portfolio—typically interpreted as 60 percent equities and 40 percent high-quality bonds—is widely used as a strategic reference point. Its appeal is understandable. Equities supply long-term growth potential, while bonds can provide income, liquidity, and protection in disinflationary recessions. The allocation is also transparent and inexpensive to implement. Yet the approach depends heavily on the interaction between stock and bond returns. When inflation is stable or declining, weak growth may lead to falling yields and positive bond returns, offsetting equity losses. When inflation is elevated and monetary policy tightens, however, rising discount rates can pressure both equities and nominal bonds. Under that regime, the same portfolio can become less diversified precisely when protection is most valuable.

This study tests whether a broader, regime-aware allocation offered a better balance between growth and preservation for a hypothetical high-net-worth investor. The analysis compares six portfolios under common data, timing, rebalancing, and transaction-cost assumptions. The empirical design emphasizes what a client could have earned using information available at each decision date rather than what an optimizer could fit with hindsight.

1.1 Research Question

The primary research question is:

> **Primary question**
>
> Can a transparent, long-only, regime-sensitive multi-asset portfolio improve downside protection and risk-adjusted performance, after reasonable transaction costs, relative to a traditional 60/40 portfolio for a hypothetical high-net-worth investor?

The question has three components. First, does expanding beyond U.S. stocks and core bonds improve diversification? Second, do downside-focused construction methods such as equal-risk contribution and conditional value-at-risk minimization produce more stable risk than mean-variance optimization? Third, does a simple macroeconomic overlay add value after accounting for the turnover and model risk created by changing allocations?

1.2 Contribution

The contribution is practical. The study connects portfolio theory to a private-bank decision process by beginning with a client profile, defining measurable objectives, and comparing simple and complex solutions on growth, real return, tail loss, drawdown, recovery, and turnover.

A second contribution is the research protocol. Portfolio weights are estimated from rolling historical windows, macroeconomic signals are lagged, and all performance from 2013 onward is out of sample. The paper reports weak outcomes rather than redefining success after observing the data.

1.3 Testable Hypotheses

Table 1. Research hypotheses

Hypothesis	Testable statement	Primary evidence	Outcome
H1	A diversified multi-asset portfolio produces a smaller maximum drawdown than 60/40.	Maximum drawdown and recovery	Not supported: strategic diversified drawdown was slightly worse.
H2	ERC and minimum-CVaR improve downside-risk-adjusted performance.	Sortino, CVaR, drawdown	Partly supported: minimum-CVaR cut tail risk; ERC was weaker.
H3	Gold and listed real estate provide regime-dependent, not universal, inflation protection.	Returns by macro regime	Supported qualitatively; benefits varied across regimes.
H4	The regime-sensitive overlay improves downside protection net of costs without excessive turnover.	CAGR, Sharpe, CVaR, drawdown, turnover	Partly supported on losses, rejected overall under the success standard.
H5	60/40 remains competitive in falling-inflation environments but is vulnerable in joint stock-bond stress.	Regime and 2022 stress results	Supported.

2. Literature Review

2.1 Modern Portfolio Theory and the Role of Diversification

Markowitz (1952) formalized portfolio selection as a joint decision about expected return, variance, and covariance. The key insight is that an asset should not be judged only by its stand-alone volatility; its value also depends on how its returns interact with the rest of the portfolio. This framework established the efficient frontier and provided a mathematical explanation for diversification. Sharpe (1964) subsequently connected systematic risk to equilibrium expected returns through the capital asset pricing model. Together, these ideas remain foundational for strategic asset allocation.

The practical weakness of sample-based mean-variance optimization is estimation error. Expected returns are difficult to estimate, and small changes in inputs can lead to large changes in portfolio weights. DeMiguel, Garlappi, and Uppal (2009) show that sophisticated optimization methods often struggle to outperform the simple 1/N rule out of sample because estimated gains are offset by parameter uncertainty and turnover. Their result is directly relevant to this study: every optimized portfolio must beat simple, investable benchmarks under the same assumptions before complexity can be considered valuable.

2.2 Downside Risk and Conditional Value-at-Risk

Variance treats upside and downside deviations symmetrically, although most clients experience risk primarily as loss, inability to fund obligations, or abandonment of the plan during a severe drawdown. Value-at-Risk summarizes a loss threshold at a chosen probability level, but it does not describe the average loss beyond that threshold. Rockafellar and Uryasev (2000) develop a tractable optimization framework for Conditional Value-at-Risk (CVaR), also known as expected shortfall, which focuses on the tail of the loss distribution. A minimum-CVaR portfolio can therefore be aligned more directly with wealth-preservation objectives than a portfolio optimized solely on variance.

CVaR is not a complete description of risk. Historical tail estimates are sensitive to the sample period, and extreme events can differ from the past. Nevertheless, it provides a useful complement to volatility and maximum drawdown because it measures the average severity of bad outcomes rather than only their frequency.

2.3 Risk Budgeting and Equal-Risk Contribution

Capital weights do not reveal risk concentration. In a conventional balanced portfolio, equities can contribute a much larger share of total volatility than their capital weight implies. Maillard, Roncalli, and Teïletche (2010) examine portfolios in which each component contributes equally to total risk. Equal-risk-contribution allocation does not require expected-return estimates, making it potentially more robust than mean-variance optimization. Its main limitation is that low-volatility assets can receive large capital weights, and a fully risk-balanced portfolio may require leverage to reach a desired return. Because leverage is inconsistent with the hypothetical client's conservative implementation constraints, this paper uses an unlevered, long-only version.

2.4 Regime-Sensitive Asset Allocation

Asset returns, volatilities, and correlations are not constant through time. Ang and Bekaert (2002) show that international return distributions can shift between regimes and that correlations may increase during volatile bear markets. This creates a challenge for static diversification: relationships estimated in calm periods may weaken during stress. A regime-sensitive approach attempts to condition allocations on observable economic information, but it also introduces classification error and trading costs. The relevant question is not whether regimes exist in hindsight; it is whether a simple rule can identify useful states in real time without overfitting.

The present design uses inflation and monetary policy rather than a hidden statistical state model. This choice sacrifices some flexibility in exchange for transparency. A private client and an investment committee can understand why the portfolio changes when inflation is rising and policy is tightening. The rule can also be reconstructed from publicly available data and audited for look-ahead bias.

2.5 Inflation Hedging Across Asset Classes

Inflation protection is more complex than assigning a permanent label such as "real asset." Roache and Attié (2009) argue that hedging effectiveness depends on the investment horizon, while Brière and Signori (2011) emphasize that economic regimes matter for inflation-hedging portfolios. More recent research by Fang, Liu, and Roussanov (2022) distinguishes core inflation from energy inflation and finds that assets commonly described as inflation hedges may behave differently across inflation components. These findings warn against assuming that gold, commodities, equities, or real estate will consistently offset every type of inflation shock.

For this reason, the paper does not define success as a positive average correlation with headline inflation. It evaluates real returns, drawdowns, and tail losses across inflation-policy states. An asset can contribute to a portfolio even when it is not a pure inflation hedge, provided its behavior improves the client-level outcome.

2.6 Implications for Private Wealth Management

Academic portfolio research often evaluates a representative investor with a single-period objective. Private wealth management adds practical constraints: liquidity needs, taxes, legacy objectives, behavioral tolerance, account structure, concentration in private business or real estate, and the requirement that recommendations remain understandable. A strategy with a slightly higher backtested Sharpe ratio may be unsuitable if it is difficult to explain, requires frequent trading, or creates a drawdown pattern the client will not tolerate. The investment-policy statement in the next section converts these concerns into explicit rules.

3. Hypothetical Client Profile and Investment Policy Statement

The investor is a fictional household created for research purposes. The profile is not based on a real client and is not a recommendation to any individual. The objective is to provide a consistent decision context against which the portfolios can be evaluated.

Table 2. Hypothetical investment-policy profile

Dimension	Assumption	Portfolio implication
Investable assets	$5,000,000	Sufficient scale for broad ETF implementation; no need for leverage or illiquid products.
Time horizon	10 years or longer	Supports meaningful equity exposure but does not eliminate sequence-of-returns risk.
Risk tolerance	Moderate	The client accepts market fluctuation but prioritizes avoidance of severe permanent loss.
Return objective	CPI + 4% annualized over a full cycle, net of estimated trading costs	Targets real growth while recognizing that the objective is not guaranteed.
Liquidity need	$100,000 annually, equivalent to 2% of initial assets	Requires liquid holdings and a minimum cash-equivalent allocation.
Maximum tolerable drawdown	Approximately 20% as an ex ante planning threshold	Portfolios exceeding the threshold require explicit justification and stress analysis.
Legal/regulatory	U.S.-based taxable investor; no leverage, short selling, derivatives, or private funds in baseline	Keeps the study implementable and transparent.
Tax treatment	Modeled only qualitatively	Pre-tax backtest results must not be presented as after-tax client outcomes.
Legacy objective	Preserve substantial principal for future generations	Favors resilience, liquidity, and disciplined rebalancing.

3.1 Return Objective

The long-term objective is to preserve purchasing power and compound real wealth. A CPI-plus target is more appropriate than a fixed nominal return because the client's future spending and legacy goals are affected by inflation. The target of CPI plus 4 percent is an aspiration, not a guaranteed outcome. The completed analysis evaluates the objective using full-period real CAGR together with drawdown, CVaR, and recovery measures.

3.2 Risk Objective

Risk is defined in several dimensions. Volatility captures the variability of returns. Maximum drawdown measures the peak-to-trough loss. CVaR estimates the average loss in the worst five percent of monthly outcomes. Recovery time measures how long the portfolio remains below its prior high. Real return shortfall measures failure to preserve purchasing power. No single metric is treated as decisive; the recommendation must be supported by a consistent pattern across measures.

3.3 Constraints

All portfolios are long only, fully invested, and unlevered. Optimized portfolios cap each asset at 40 percent, require at least 5 percent in BIL, cap combined U.S. and developed-market equities at 70 percent, and cap combined listed real estate and gold at 30 percent. The analysis does not impose a turnover ceiling, but reports turnover and deducts transaction costs.

4. Data and Asset-Class Proxies

4.1 Sample Window and Frequency

The dataset runs from January 2008 through December 2025 at month-end frequency. The first 60 months are reserved for estimation, so the principal out-of-sample comparison begins in January 2013 and contains 156 monthly observations.

ETF market data were retrieved on July 17, 2026 at 02:28:20 UTC using yfinance version 1.5.1 with automatic price adjustment enabled. The raw retrieval window begins in January 2007, while the research sample begins in January 2008. Daily adjusted prices were converted to month end observations and simple monthly total returns. VEA has six initial missing month end observations and BIL has four because those funds did not yet have complete trading histories at the beginning of the raw retrieval window. These pre inception gaps occur before the January 2008 research sample and do not represent deleted observations. CPIAUCSL, FEDFUNDS, and USREC were retrieved from FRED using pandas datareader version 0.11.1. Data processing used pandas version 3.0.3, and the complete Python environment is recorded in requirements.txt

Table 3. Investable proxies

Exposure	Proxy	Role in portfolio	Primary risk driver
U.S. equities	SPY - SPDR S&P 500 ETF Trust	Long-term growth and U.S. corporate earnings	Growth, valuation, equity risk premium
Developed ex-U.S. equities	VEA - Vanguard FTSE Developed Markets ETF	Geographic and currency diversification	Global growth, currency, regional valuation
Intermediate Treasuries	IEF - iShares 7-10 Year Treasury Bond ETF	Duration exposure and recession defense	Nominal yields, inflation expectations, term premium
Investment-grade credit	LQD - iShares iBoxx $ Investment Grade Corporate Bond ETF	Income and credit-spread exposure	Treasury yields, credit spreads, default risk
Listed real estate	VNQ - Vanguard Real Estate ETF	Real-asset and property-income exposure	Rates, property fundamentals, equity beta
Gold	GLD - SPDR Gold Shares	Crisis and inflation diversification	Real rates, dollar, risk aversion
Treasury bills / cash	BIL - SPDR Bloomberg 1-3 Month T-Bill ETF	Liquidity and risk-free proxy	Short-term policy rates

4.2 Macroeconomic Variables

Headline inflation is measured using CPIAUCSL, and monetary-policy direction is measured using FEDFUNDS. To prevent look-ahead bias, the regime strategy uses a two-month signal lag. Because BLS did not publish a complete October 2025 CPI index during the 2025 federal shutdown, the strategy retains the latest genuinely available inflation observation rather than interpolating a new index value.

Table 4. Macroeconomic data architecture

Variable	Series/source	Transformation	Use
Headline inflation	CPIAUCSL, FRED/BLS	Year-over-year change; two-month signal lag; last published value retained when a release is unavailable	Inflation trend and real-growth measurement
Policy rate	FEDFUNDS, FRED	Six-month change after a two-month signal lag	Tightening versus easing classification
Cash / risk-free proxy	BIL adjusted total return	Monthly ETF total return	Sharpe ratio and portfolio cash sleeve
Recession indicator	USREC, FRED / NBER chronology	Monthly 0/1 indicator	Stress-period reporting

4.3 Return Construction and Data Cleaning

Adjusted prices were converted to simple monthly total returns. All series were aligned to the final available trading date of each calendar month. Missing observations were not forward-filled across a month without documentation. The common sample began only after every ETF proxy had a valid return, and the validation report recorded missing values, duplicate dates, non-month-end dates, and extreme-return checks.

$$r_{i,t} = (P_{i,t} \,/\, P_{i,t-1}) - 1 \tag{1}$$

Real CAGR is calculated by dividing cumulative nominal portfolio growth by the cumulative CPI growth factor between December 2012 and December 2025, then annualizing the resulting real wealth ratio. This endpoint method avoids inventing a missing monthly October 2025 CPI observation.

$$r_{real,t} = [(1 + r_{p,t}) \,/\, (1 + \pi_t)] - 1 \tag{2}$$

4.4 Data Governance and Reproducibility

The public repository contains the six analysis scripts, a pinned requirements.txt file, project documentation, sanitized retrieval metadata, validation reports, portfolio diagnostics, and the authoritative result tables and figures used in Working Paper 1.0. The reported results correspond to the data retrieval completed on July 17, 2026 at 02:28:20 UTC. Raw third-party market-price files are not redistributed. The data collection script permits an independent refresh of the dataset, although later downloads may differ because third party providers can revise historical adjusted-price information. Files prefixed fair_comparison_ contain the authoritative six portfolio results reported in this paper, files prefixed baseline_ and optimized_ are intermediate audit outputs. The retrieval log is stored at data/metadata/data_retrieval_log.txt. The repository is available at https://github.com/ishaanparm/beyond-60-40-portfolio.

5. Methodology

5.1 Portfolio Set

Table 5. Portfolio definitions

Portfolio	Construction rule	Purpose
Traditional 60/40	60% SPY and 40% IEF; quarterly rebalance	Simple balanced benchmark
Strategic diversified	35% SPY, 15% VEA, 15% IEF, 10% LQD, 10% VNQ, 10% GLD, 5% BIL	Fixed multi-asset benchmark
Constrained mean-variance	Rolling 60-month maximum historical Sharpe portfolio under policy limits	Tests return-risk optimization with guardrails
Equal-risk contribution	Weights selected to equalize variance-contribution shares under the same limits	Tests risk budgeting without leverage
Minimum CVaR	Historical 95% CVaR minimization under the same limits	Tests explicit tail-loss control
Regime-sensitive	One of four fixed allocations selected using lagged inflation and policy trends	Tests transparent tactical allocation

5.2 Rebalancing Convention

All portfolios are evaluated monthly. Fixed and optimized portfolios rebalance before the January, April, July, and October returns are earned. Optimized weights use only the preceding 60 months. The first allocation is not charged a cost; later reallocations incur 10 basis points multiplied by one-way turnover.

$$r_{p,t} = \Sigma_i \, w_{i,t-1} \, r_{i,t} \tag{3}$$

Turnover at a rebalance is one half of the sum of absolute changes in weights. The factor of one half avoids double-counting the sale and purchase sides of the same reallocation. Net return deducts a baseline cost of 10 basis points per unit of one-way turnover. Transaction-cost sensitivity was evaluated using 0, 10, and 25 basis points per unit of one way turnover.

$$\text{Turnover}_t = 0.5 \times \Sigma_i \, |w_{i,t}{}^{*} - w_{i,t}{}^{-}| \tag{4}$$

$$r_{p,t}{}^{net} = r_{p,t}{}^{gross} - c \times \text{Turnover}_t \tag{5}$$

5.3 Constrained Mean-Variance Portfolio

The mean-variance portfolio uses the preceding 60 monthly returns to maximize the historical Sharpe ratio, with the mean BIL return in the estimation window serving as the cash benchmark. The model is solved using constrained nonlinear optimization and does not use future observations.

$$\max\_w \ [(w'\hat{\mu} - \hat{r}\_f) / \sqrt{(w'\hat{\Sigma}w)}] \tag{6}$$

Constraints are: weights sum to one; all weights are nonnegative; each asset is capped at 40 percent; BIL is at least 5 percent; SPY plus VEA is no more than 70 percent; and VNQ plus GLD is no more than 30 percent.

Because sample expected returns are unstable, the paper treats corner solutions and repeated binding constraints as important diagnostics rather than evidence of a naturally diversified optimum.

5.4 Equal-Risk-Contribution Portfolio

For portfolio volatility $\sigma_p = \sqrt{(w'\Sigma w)}$, the marginal contribution of asset i is $(\Sigma w)_i / \sigma_p$. The total risk contribution is the asset weight multiplied by its marginal contribution. The ERC portfolio chooses weights so that each asset contributes approximately one seventh of total volatility, subject to the same long-only, concentration, equity, and cash constraints.

$$RC_i = w_i(\Sigma w)_i / \sigma_p \tag{7}$$

$$RC_i \approx \sigma_p / N \text{ for all i} \tag{8}$$

Because the portfolio is not levered, equal risk contribution may produce a lower expected return than an institutional risk-parity portfolio that scales exposures. That is intentional: leverage would change the risk profile and complicate comparison with the client mandate.

5.5 Minimum-CVaR Portfolio

The minimum-CVaR portfolio minimizes historical monthly 95 percent CVaR over the rolling 60-month window using linear programming. The same long-only and policy constraints apply.

$$CVaR_{0.95}(L) = E[L \mid L \geq VaR_{0.95}(L)] \tag{9}$$

No minimum expected-return constraint is imposed. The resulting portfolio is therefore interpreted as a defensive tail-risk solution rather than a return-maximizing allocation.

5.6 Regime Classification

The regime model uses observable, lagged macroeconomic variables. Inflation is classified as rising or falling according to the six-month change in the two-month-lagged year-over-year CPI rate. Policy is classified as tightening or easing according to the six-month change in the two-month-lagged effective federal funds rate. Zero changes retain the previous state.

$$\text{InflationState}_t = \text{Rising if } \Delta_6[\text{YoY CPI}_{t-2}] > 0\text{; otherwise Falling} \tag{10}$$

$$\text{PolicyState}_t = \text{Tightening if } \Delta_6[\text{FEDFUNDS}_{t-2}] > 0\text{; otherwise Easing} \tag{11}$$

Table 6. Precommitted regime-sensitive allocation matrix

Regime	SPY	VEA	IEF	LQD	VNQ	GLD	BIL	Economic rationale
Rising inflation / Tightening	20%	5%	10%	10%	10%	20%	25%	Defensive liquidity and gold; lower equity and duration exposure.
Rising inflation / Easing	30%	10%	10%	10%	15%	15%	10%	Moderate growth and real assets with retained liquidity.
Falling inflation / Tightening	25%	10%	20%	15%	5%	5%	20%	Quality fixed income and cash while policy remains restrictive.
Falling inflation / Easing	35%	15%	20%	10%	10%	5%	5%	Higher growth and duration exposure in disinflationary easing.

The allocation matrix was fixed before the final backtest. It is intentionally coarse and does not claim to forecast markets. Its value was judged by net performance, drawdown reduction, and the precommitted decision standard.

5.7 Out-of-Sample Design

The main test uses a rolling 60-month estimation window. January 2008 through December 2012 is used only for the first set of estimates, and reported optimized performance begins in January 2013. Every quarterly rebalance uses data strictly earlier than the return month.

Robustness tests vary the estimation window, rebalancing frequency, maximum asset weight, and transaction cost. Stress-period tests separately report the COVID shock, full-year 2020, the 2022 rate shock, and NBER recession months.

5.8 Performance Measures

Table 7. Evaluation metrics

Metric	Definition	Client relevance
CAGR	Geometric annualized growth rate	Long-term compounding of nominal wealth
Real CAGR	CAGR after CPI adjustment	Purchasing-power preservation
Annualized volatility	Monthly standard deviation × √12	Variability and planning uncertainty
Sharpe ratio	Annualized excess return / annualized volatility	Return per unit of total variability
Sortino ratio	Annualized excess return / downside deviation	Return relative to harmful volatility
Maximum drawdown	Largest peak-to-trough decline	Behavioral and capital-preservation stress
Recovery time	Months from prior peak to a new high	Duration of financial and behavioral pressure
95% VaR	Fifth percentile monthly return expressed as loss	Loss threshold under historical simulation
95% CVaR	Average loss in worst 5% of months	Severity of tail outcomes
Turnover	Weight traded at each rebalance	Implementation cost and tax sensitivity
Final wealth of $1	Cumulative value of one initial dollar after net returns	Communicates the realized compounding outcome

5.9 Implementation Diagnostics

The analysis inspects optimized weights, binding constraints, turnover, and solver diagnostics. The maximum-Sharpe model frequently reaches the 40 percent limits in SPY, IEF, or BIL and assigns near-zero weights to weaker estimated assets, illustrating the sensitivity of sample mean-variance optimization.

5.10 Robustness Tests

- Estimation windows of 36, 60, and 84 months.
- Monthly, quarterly, and semiannual rebalancing.
- Transaction costs of 0, 10, and 25 basis points per unit of one way turnover.
- A stricter 30 percent maximum weight for each asset.
- COVID shock, full-year 2020, 2022 rate shock, and NBER recession month tests.
- Multiple feasible starting allocations were used in the robustness analysis to reduce sensitivity to numerical initialization. The primary 2013–2025 mean variance model retained its original optimization specification.

6. Empirical Results

6.1 Main Out-of-Sample Performance

The principal comparison covers 156 months from January 2013 through December 2025. All figures are net of 10 basis points per unit of one-way turnover. The traditional 60/40 portfolio produced the strongest compounding, while the optimized portfolios delivered substantially lower volatility and drawdown.

Table 8. Net out-of-sample portfolio performance, 2013-2025

Portfolio	CAGR	Real CAGR	Vol.	Sharpe	Sortino	Max DD	95% CVaR	Turnover
Traditional 60/40	9.56%	6.70%	9.08%	0.885	1.391	-20.56%	5.21%	6.21%
Strategic diversified	8.58%	5.75%	9.37%	0.766	1.187	-20.94%	5.51%	6.93%
Mean-variance (Max Sharpe)	6.56%	3.78%	4.94%	1.036	1.734	-12.16%	2.68%	23.11%
Equal risk contribution	5.02%	2.28%	6.04%	0.595	0.954	-15.21%	3.34%	41.76%
Minimum CVaR	3.99%	1.28%	3.96%	0.634	0.987	-12.78%	2.27%	12.96%
Regime-sensitive	6.80%	4.02%	7.88%	0.683	1.057	-17.44%	4.68%	43.02%

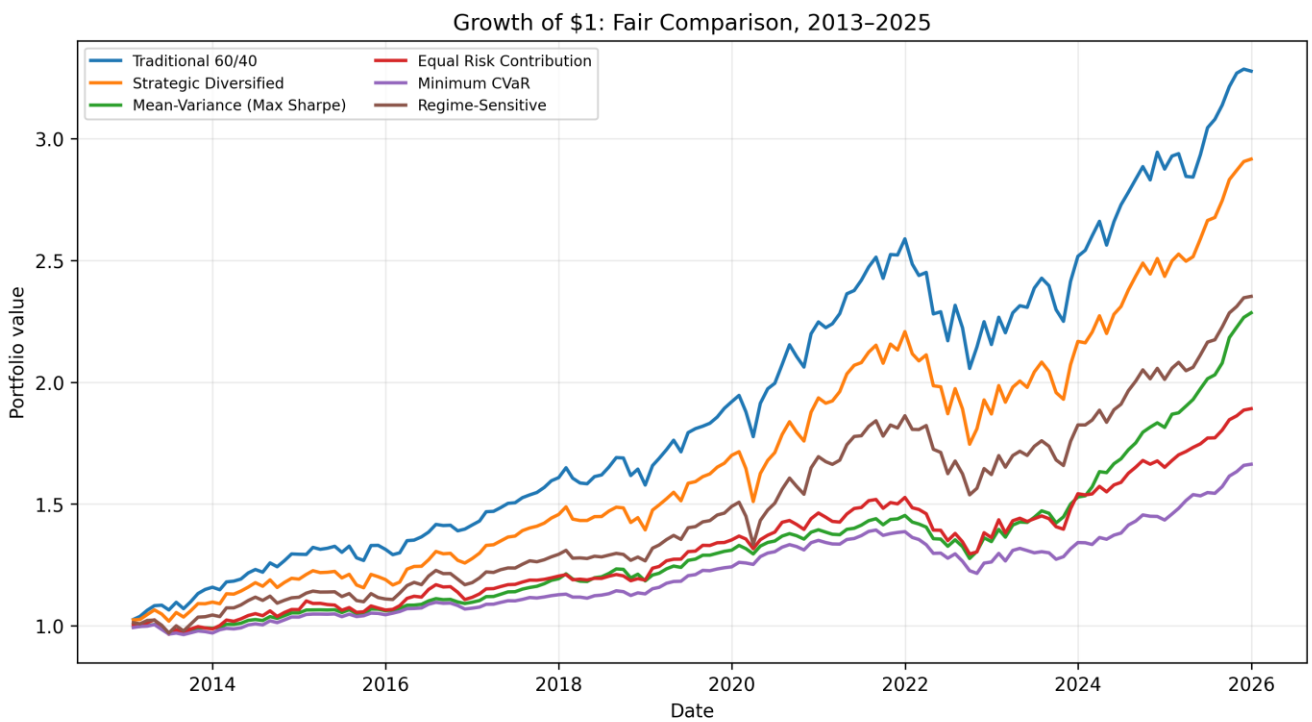

Figure 1. Growth of $1 for all six portfolios, January 2013-December 2025.

The 60/40 portfolio ended with $3.28 for each initial dollar, compared with $2.35 for the regime-sensitive portfolio and $2.28 for mean-variance. The growth advantage of 60/40 was especially visible after 2023, while the defensive portfolios compounded more slowly throughout most of the sample.

6.2 Drawdown and Stress-Period Results

The largest drawdowns occurred during the 2022 joint decline in stocks and bonds. Mean-variance and minimum-CVaR sharply reduced peak-to-trough losses. Minimum-CVaR, however, remained below a prior peak for longer because its return recovery was slow.

Table 9. Maximum drawdown episodes

Portfolio	Max DD	Peak	Trough	Recovery	Longest DD (mo.)
Traditional 60/40	-20.56%	Dec. 2021	Sep. 2022	Feb. 2024	25
Strategic diversified	-20.94%	Dec. 2021	Sep. 2022	Mar. 2024	26
Mean-variance	-12.16%	Dec. 2021	Sep. 2022	Jul. 2023	18
Equal risk contribution	-15.21%	Dec. 2021	Sep. 2022	Dec. 2023	23
Minimum CVaR	-12.78%	Aug. 2021	Oct. 2022	Jul. 2024	34
Regime-sensitive	-17.44%	Dec. 2021	Sep. 2022	Mar. 2024	26

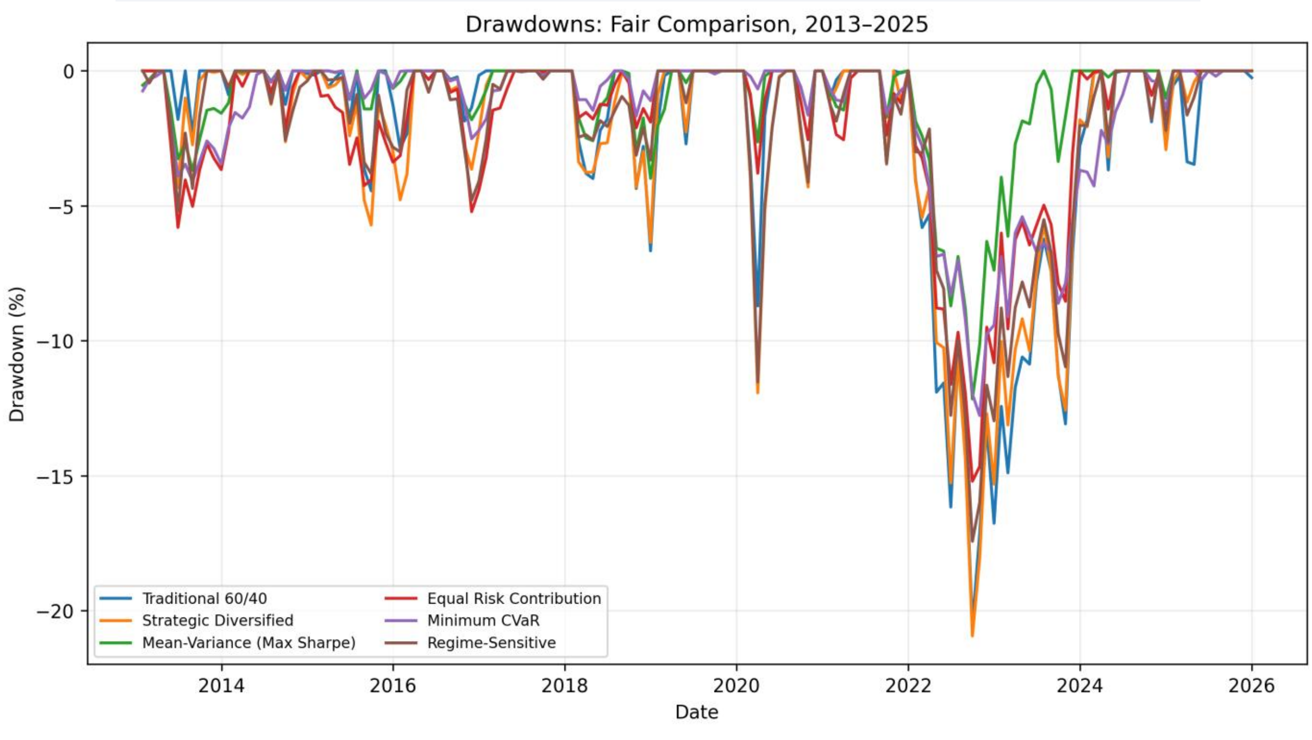

Figure 2. Portfolio drawdowns, January 2013-December 2025.

Table 10. Cumulative returns during selected stress periods

Stress period	60/40	Strategic	Mean-var.	ERC	Min-CVaR	Regime
COVID shock (Feb.-Mar. 2020)	-8.71%	-11.93%	-2.63%	-3.79%	-0.67%	-11.53%
Full-year 2020	17.08%	13.88%	6.39%	8.18%	8.75%	13.71%
2022 rate shock	-16.77%	-15.32%	-7.39%	-10.82%	-8.99%	-12.97%
NBER recession months	1.87%	-1.21%	0.82%	-0.20%	2.08%	-1.18%

Minimum-CVaR was the most protective portfolio during the two-month COVID shock and generated the best cumulative return during the NBER recession months. Mean-variance also limited losses effectively in 2020 and 2022. The regime-sensitive portfolio did not protect well during the abrupt COVID shock because the lagged macro classification could not anticipate the event.

6.3 Performance by Inflation and Policy Regime

The sample contained 62 rising-inflation/tightening months, 21 rising-inflation/easing months, 45 falling-inflation/tightening months, and 28 falling-inflation/easing months. The table reports annualized returns within each state; these conditional rates should not be interpreted as independent long-horizon forecasts.

Table 11. Annualized return by macroeconomic regime

Regime	Months	60/40	Strategic	Mean-var.	ERC	Min-CVaR	Regime
Rising inflation / Tightening	62	5.60%	3.50%	2.09%	1.09%	0.46%	1.34%
Rising inflation / Easing	21	6.46%	6.69%	7.50%	5.79%	7.80%	6.65%
Falling inflation / Tightening	45	11.12%	11.04%	10.62%	8.56%	6.43%	8.95%
Falling inflation / Easing	28	18.74%	18.02%	9.64%	7.78%	5.28%	16.30%

The regime sensitive portfolio produced a strong 16.30 percent annualized return during falling inflation/easing months, although it remained below the 60/40 portfolio's 18.74 percent return. It slightly outperformed 60/40 during rising-inflation/easing months, returning 6.65 percent compared with 6.46 percent. It underperformed 60/40 during rising-inflation/tightening and falling-inflation/tightening months.

6.4 Transaction-Cost Sensitivity

Changing the one-way turnover cost from 0 to 25 basis points produced only small changes in CAGR and did not alter the portfolio ranking. The effect was largest for ERC and the regime-sensitive strategy because they had the highest turnover.

Table 12. CAGR under alternative transaction costs

Portfolio	0 bps	10 bps	25 bps	25-bps Sharpe	25-bps Max DD
Traditional 60/40	9.57%	9.56%	9.55%	0.884	-20.56%
Strategic diversified	8.59%	8.58%	8.57%	0.765	-20.95%
Mean-variance	6.59%	6.56%	6.53%	1.029	-12.23%
Equal risk contribution	5.07%	5.02%	4.96%	0.585	-15.27%
Minimum CVaR	4.00%	3.99%	3.97%	0.629	-12.85%
Regime-sensitive	6.85%	6.80%	6.73%	0.676	-17.49%

6.5 Mean-Variance Robustness

For a common January 2015-December 2025 test window, the maximum-Sharpe result remained broadly stable across alternative estimation windows, rebalancing frequencies, and a stricter asset cap. The range of maximum drawdowns was 11.41 to 13.80 percent.

Table 13. Mean-variance parameter robustness, 2015-2025

Variant	CAGR	Vol.	Sharpe	Max DD	95% CVaR	Turnover
36-month lookback	6.83%	5.66%	0.899	-12.48%	2.84%	38.11%
Main 60-month	7.30%	5.19%	1.067	-12.16%	2.76%	21.67%
84-month lookback	6.98%	5.33%	0.981	-12.35%	2.82%	17.30%
Monthly rebalance	7.13%	5.20%	1.035	-13.80%	2.94%	37.11%
Semiannual rebalance	7.41%	5.16%	1.092	-11.41%	2.67%	14.78%
30% asset cap	7.75%	5.69%	1.045	-13.02%	3.05%	17.09%

The robustness results do not establish that the model will perform similarly in live markets, but they reduce the concern that the main downside result arose from a single arbitrary lookback or rebalance schedule. The semiannual variant had the highest Sharpe ratio and the smallest drawdown, while the 30 percent cap produced the highest CAGR.

6.6 Precommitted Decision Standard

The regime-sensitive portfolio had to satisfy all four criteria to be considered economically superior. It did not.

Table 14. Evaluation of the regime-sensitive strategy

Criterion	Evidence	Result
Drawdown improvement	17.44% versus 20.56%; 3.12 percentage points and 15.2% relative improvement	Pass
Lower 95% CVaR	4.68% versus 5.21%	Pass
Real CAGR within 1 point and better Sharpe or Sortino	4.02% versus 6.70%; Sharpe 0.683 versus 0.885	Fail
Turnover below 100% and robust qualitative conclusion	43.02% turnover; cost sensitivity did not change ranking	Pass on turnover; limited rule robustness
Overall standard	All conditions were required	Fail

Because the real-return and risk-adjusted-return condition failed, the additional tactical complexity was not justified for the hypothetical moderate-risk client. The strategy may still suit a client who explicitly accepts lower growth in exchange for partial drawdown reduction.

7. Discussion

7.1 Simplicity and the Strength of the Benchmark

The 60/40 portfolio won on both nominal and real growth despite experiencing a larger drawdown than the optimized portfolios. Its advantage came from maintaining substantial equity participation while high-quality Treasuries continued to diversify many, though not all, shocks. The result cautions against assuming that a larger number of asset classes automatically creates a superior portfolio.

7.2 Optimization as a Defensive Tool

The mean-variance portfolio generated the strongest risk-adjusted performance and recovered from the 2022 drawdown more quickly than the benchmark. Yet its lower CAGR and repeated boundary weights show that optimization solved a different problem: it prioritized historical excess return per unit of volatility rather than maximum wealth accumulation. Minimum-CVaR was even more defensive and produced the lowest tail loss, but its real CAGR of 1.28 percent was too low for the stated CPI-plus-four objective.

7.3 Why the Regime Strategy Fell Short

The regime strategy reduced drawdown and CVaR but lagged in three of the four macro states. Its predetermined weights were deliberately simple, and the two-month signal lag protected against look-ahead bias; those same features made the strategy slow to respond to abrupt shocks. The result does not prove that all regime methods fail. It shows that this transparent specification did not earn enough return to compensate for its turnover and tactical complexity.

7.4 Client Trade-Offs and Governance

A private-bank recommendation should begin with the client objective. A growth-oriented client able to tolerate a roughly 20 percent drawdown would have preferred the simple benchmark in this sample. A client more concerned with sequence risk or behavioral abandonment could rationally accept the mean-variance portfolio's lower return in exchange for materially lower volatility and drawdown. The choice is therefore an investment-policy decision, not a mechanical ranking exercise.

8. Private Wealth Management Implications

8.1 Suitability and Client Discovery

The hypothetical allocation is not suitable for every wealthy investor. A real advisory process would examine the client's operating business, concentrated stock, real estate, employment income, debt, tax basis, trusts, charitable plans, and family governance. An entrepreneur whose net worth is already tied to an illiquid business may need a more defensive liquid portfolio than the standalone investment account suggests. Conversely, a client with stable pension income and no near-term distributions may tolerate more equity risk.

8.2 Liquidity Segmentation

One implementation approach is to segment the portfolio into liquidity, stability, and growth pools. The liquidity pool funds several years of expected withdrawals using Treasury bills and short-duration instruments. The stability pool contains high-quality bonds and selected diversifiers. The growth pool contains global equities and listed real estate. Although the backtest evaluates a single combined portfolio, this segmentation can improve communication because each asset has a stated job rather than a generic return target.

8.3 Tax and Account Location

The baseline study is pre-tax. In practice, bond interest, REIT distributions, capital gains, and rebalancing can have different tax consequences. Asset location across taxable accounts, retirement accounts, trusts, and charitable vehicles may materially affect the after-tax result. High-turnover tactical strategies face a higher burden of proof in taxable portfolios. Therefore, the final recommendation should report turnover and explicitly state that after-tax optimization is outside the paper's empirical scope.

8.4 Product Selection and Due Diligence

The ETF proxies are research instruments, not endorsements. A real implementation would evaluate tracking difference, bid-ask spreads, assets under management, securities lending, tax structure, duration, credit quality, and operational risk. The portfolio decision should be separated from product selection so that a conclusion about an asset class is not confused with the performance of a single fund.

8.5 Communication Standard

A private-bank recommendation should be explainable in a short client conversation. The proposed communication standard is: identify the client goal, show the role of each sleeve, explain the principal risks, provide a severe but plausible stress outcome, and state the rebalancing discipline. Complex model language should remain in the technical appendix. The client should understand what could cause the portfolio to disappoint before capital is allocated.

9. Limitations

First, the main out-of-sample period contains only 156 months. It includes the COVID shock, the 2022 joint stock-bond decline, and multiple inflation and policy states, but it does not represent every possible market environment.

Second, historical means, covariances, and tail losses are unstable. The maximum-Sharpe optimizer frequently produced boundary allocations, and live performance may differ materially from the backtest.

Third, the regime rule uses only inflation direction and policy direction. It ignores valuation, credit spreads, growth surprises, and market-implied expectations. Alternative definitions could generate different allocations.

Fourth, the analysis is pre-tax and excludes advisory fees, bid-ask spreads beyond the modeled turnover cost, investor-specific tax lots, market impact, and security-lending differences.

Fifth, ETF prices came from a convenience data source rather than a licensed institutional feed. In addition, BLS did not publish a complete October 2025 CPI index because of the 2025 federal shutdown. Real CAGR therefore uses beginning and ending CPI levels, and the regime signal carries forward the latest published inflation observation without creating a synthetic index value.

Finally, the NBER recession stress test contains only the two recession months in the 2013-2025 sample, and several portfolios and robustness variants are examined. The precommitted success rule and disclosure of weak outcomes reduce, but do not eliminate, data-mining risk.

10. Conclusion

This paper compared six liquid, long-only portfolios under a common out-of-sample design from January 2013 through December 2025. The traditional 60/40 portfolio produced the highest nominal and real growth. The constrained mean-variance portfolio delivered the strongest Sharpe and Sortino ratios and materially smaller drawdowns, while minimum-CVaR achieved the lowest volatility and tail loss at the cost of weak growth.

The precommitted regime-sensitive portfolio reduced maximum drawdown from 20.56 percent to 17.44 percent and lowered 95 percent CVaR from 5.21 percent to 4.68 percent. Those benefits were not sufficient. Its real CAGR lagged the benchmark by 2.69 percentage points, its Sharpe ratio was lower, and turnover was substantially higher. It therefore failed the paper's precommitted economic-superiority standard.

For the hypothetical moderate-risk high-net-worth client, the evidence supports retaining a simple strategic benchmark when long-run growth is the primary goal. Constrained optimization may be appropriate when the client places greater value on drawdown control, sequence-risk reduction, and behavioral staying power. The central lesson is not that one model is universally best, but that every additional layer of complexity must earn its place through a measurable improvement in the client objective.

The paper is an independent working paper rather than peer-reviewed research. Future work should test alternative public data sources, tax-aware implementation, additional macroeconomic definitions, and longer samples using index histories rather than ETF inception-limited data.

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Appendix A. Formula and Metric Definitions

Table A1. Metric definitions

Measure	Formula / definition	Implementation note
CAGR	(Final wealth / Initial wealth)^(12 / number of months) − 1	Use geometric compounding, not arithmetic average.
Annualized volatility	Standard deviation of monthly returns × √12	Report both nominal and downside measures.
Sharpe ratio	Annualized mean excess return / annualized volatility	Use a monthly risk-free series aligned to returns.
Downside deviation	$\sqrt{[12 \times \text{mean}(\min(r_t - \text{MAR}, 0)^2)]}$	Minimum acceptable return (MAR) set to monthly risk-free rate in base case.
Sortino ratio	Annualized excess return / annualized downside deviation	Interpret with observation count and distribution limits.
Maximum drawdown	Minimum of cumulative wealth / prior running maximum − 1	Report depth and duration.
Historical VaR 95%	Negative fifth percentile of monthly return	A threshold, not average tail severity.
Historical CVaR 95%	Negative average return at or below fifth percentile	Use consistent sign convention throughout.
Real CAGR	(Nominal growth / cumulative CPI growth)^(12 / number of months) − 1	Endpoint CPI method used because October 2025 CPI is unavailable.
Turnover	0.5 × sum of absolute target-minus-pretrade weight changes	Report annualized average and maximum quarter.

Appendix B. Data Dictionary

Table B1. Processed dataset fields

Field	Type	Description
date	month-end date	Common observation date after alignment
adj_price_<ticker>	float	Month-end adjusted closing price for each ETF proxy
return_<ticker>	float	Simple monthly total return from adjusted prices
cpi_index	float	CPIAUCSL index level; October 2025 is unavailable
infl_yoy	float	Twelve-month percentage change in CPI when calculable
inflation_signal	float	Two-month-lagged year-over-year CPI, carrying the last published observation when unavailable
inflation_trend	category	Rising or falling based on the six-month signal change
fedfunds	float	Effective federal funds rate in percent
policy_6m_change	float	Six-month change in the lagged policy-rate signal
policy_state	category	Tightening or easing
regime	category	Combination of inflation and policy states
nber_recession	binary	1 in NBER contraction months; otherwise 0
net_return_<portfolio>	float	Monthly portfolio return after modeled turnover cost
turnover_<portfolio>	float	One-way turnover at scheduled rebalance dates

Appendix C. Research and Investment Disclaimer

This document is an independent educational working paper. It is not peer reviewed, investment advice, a solicitation, an offer to buy or sell securities, or a recommendation for any person. The hypothetical client is fictional. Historical and backtested results do not guarantee future performance.